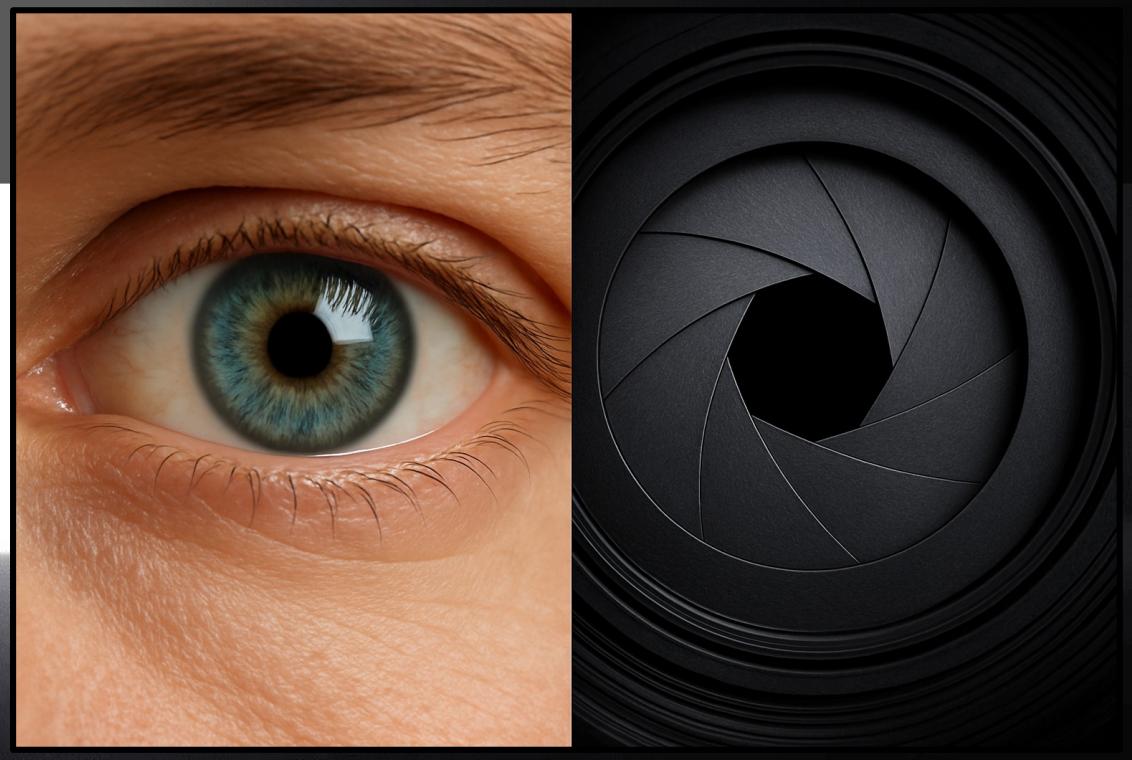
Computer Vision & NeuroVision



Prof. Hugo de Paula

What Computer Vision Aims to Achieve

01

Core Objective

Enable machines to interpret visual data like humans, transforming pixel grids into high-level understanding.

02

Challenges

Machines still struggle with adaptability and contextual reasoning compared to humans.

03

Impact

Applications span healthcare, automotive, manufacturing, and retail.

Replicating Biological Vision is Hard

01

Human Visual System

The human visual system processes information holistically, integrating prior knowledge, experience, and sensory inputs to make sense of the visual world. This adaptability and generalization are difficult to replicate in machines.

02

Machine Limitations

Machines often struggle with variations in lighting, viewpoint, and occlusion, which humans handle effortlessly. The absence of lifelong learning and top-down feedback loops in current systems exacerbates these challenges.

Recognition Reconstruction Reorganization

Reconstruction

Reconstruction focuses on creating 3D models of scenes or objects from 2D images. Techniques like stereo vision and structure from motion address the challenge.

02

Challenges

Challenges such as noise, occlusion, and the need for large datasets.

04

01

Recognition

Recognition involves identifying objects, scenes, and activities in images and videos. This includes tasks like image classification, object detection, and facial recognition.

03

Reorganization

Reorganization involves segmenting images into meaningful regions.

Pre-Digital Concepts 1700-1950

Early Developments

The early concepts of optics and photography laid the groundwork for digital imaging.

c. 1000 AD: The Camera Obscura: Ibn al-Haytham - *“Intromission theory of vision”:* identify that light enters the eye.

Kodak's first camera system in 1884

First digital photograph: Walden Kirsch, 1957

First Digital Image 1957

01

Digital Imaging

Russell Kirsch scanned a baby photo into a 176×176 binary grid on SEAC, proving photographs could become numeric data. This moment birthed digital imaging and pixel arrays.

02

Significance

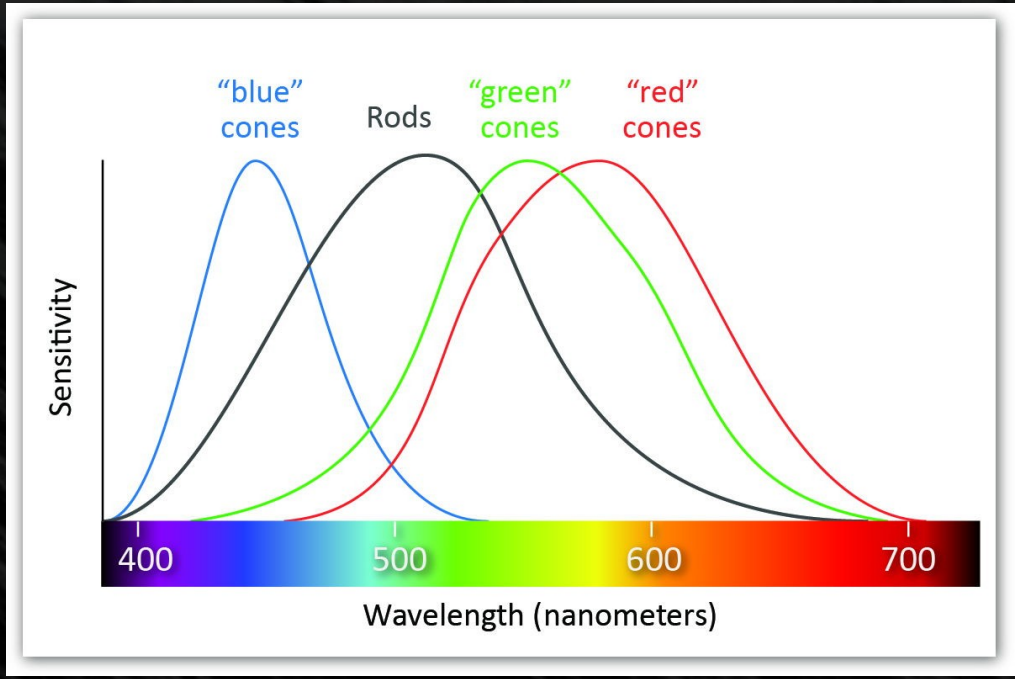
This breakthrough demonstrated that visual information could be converted into a format that computers could process.

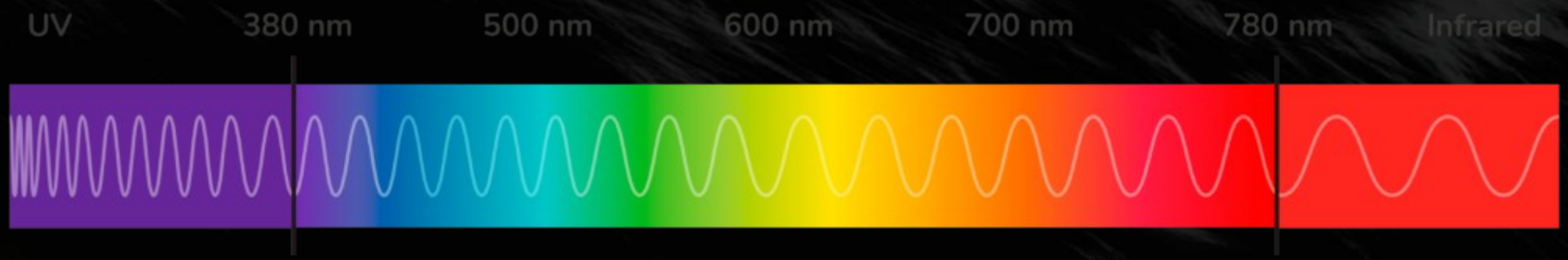
03

Future Implications

The ability to digitize images laid the foundation for the development of computer vision algorithms.

Cameras are optimized for different wavelengths





Hubel & Wiesel 1962 Nobel Insight

Hubel anad Weisel after winning their Nobel Prize, 1981.
Courtesy of Harvard University Archives.

Neuroscience Breakthrough

David H. Hubel and Torsten N. Wiesel conducted experiments on cats to better understand the natural visual system by showing a cat various patterns of light to find out when its visual cortex would start firing.

Research revealed neurons in the visual cortex selective for edge orientation and hierarchical composition. Their findings inspired later CNN architectures mimicking simple-to-complex cell layers for progressive feature extraction.

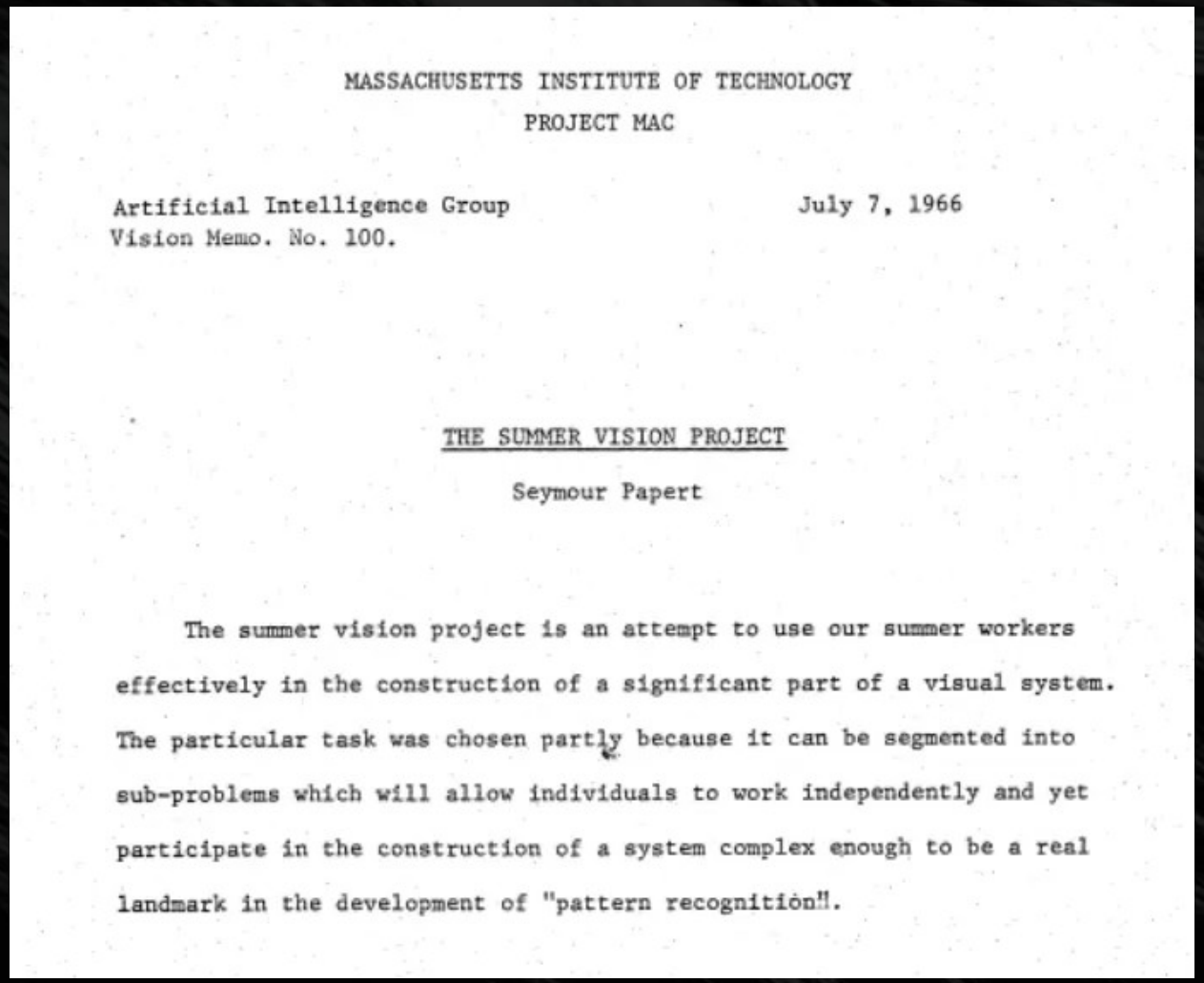
MIT Summer Vision Project 1966

Project objective

The MIT Summer Vision Project aimed to build a system that could segment and classify images.

Results

The project demonstrated that a more systematic and principled approach was needed to tackle the challenges of computer vision.



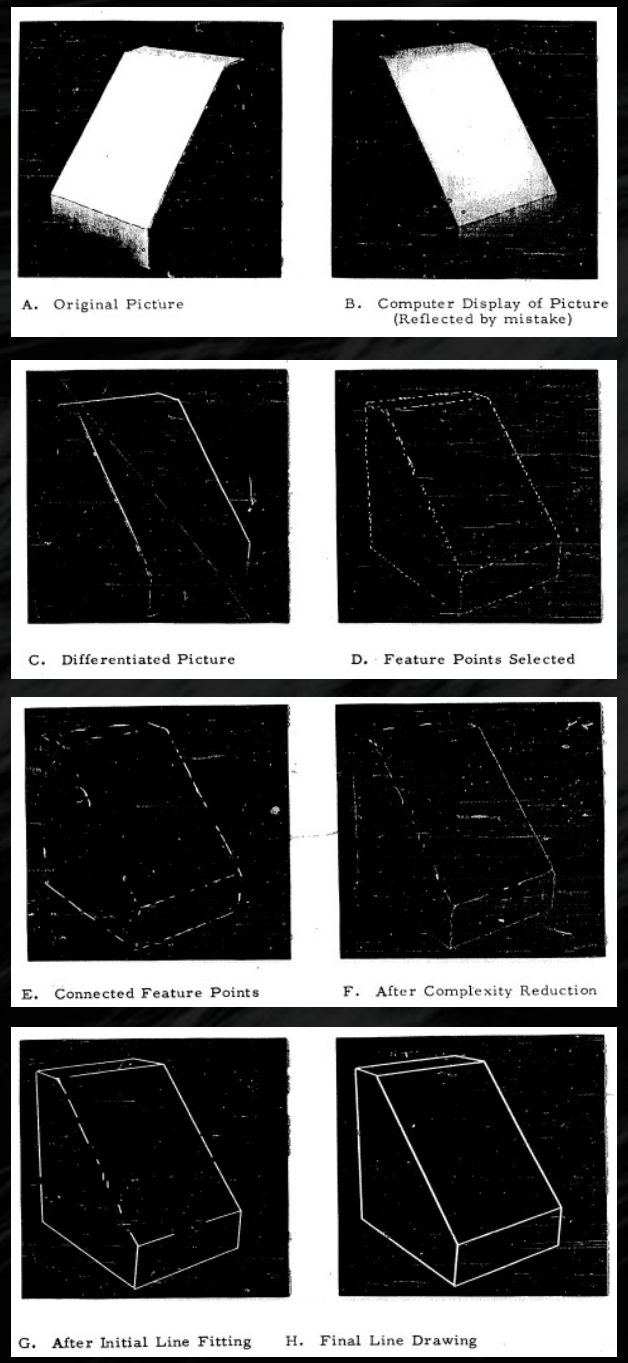
MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group — July 7, 1966
Vision Memo. No. 100.

THE SUMMER VISION PROJECT

Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".



Larry Roberts 3D Thesis 1963

3D Reconstruction

Larry Roberts showed that polyhedral objects could be reconstructed from line drawings, introducing geometric primitives and camera projection math. His work laid the foundation for 3D vision and CAD model understanding.

Legacy

Roberts' thesis introduced the 'blocks world' paradigm, which dominated early 3D vision research and inspired future developments in computer graphics and robotics.

Feature Era 1970-1990

01

Feature-Based Methods

Techniques like the Hough transform, Canny edge detection, and SIFT (Scale-Invariant Feature Transform) for robust feature extraction and matching.

02

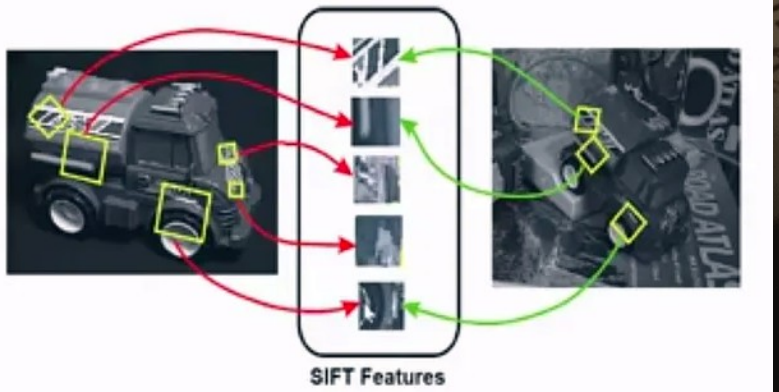
Advancements

Enabled more sophisticated image analysis: object recognition, stereo matching, and image stitching.

03

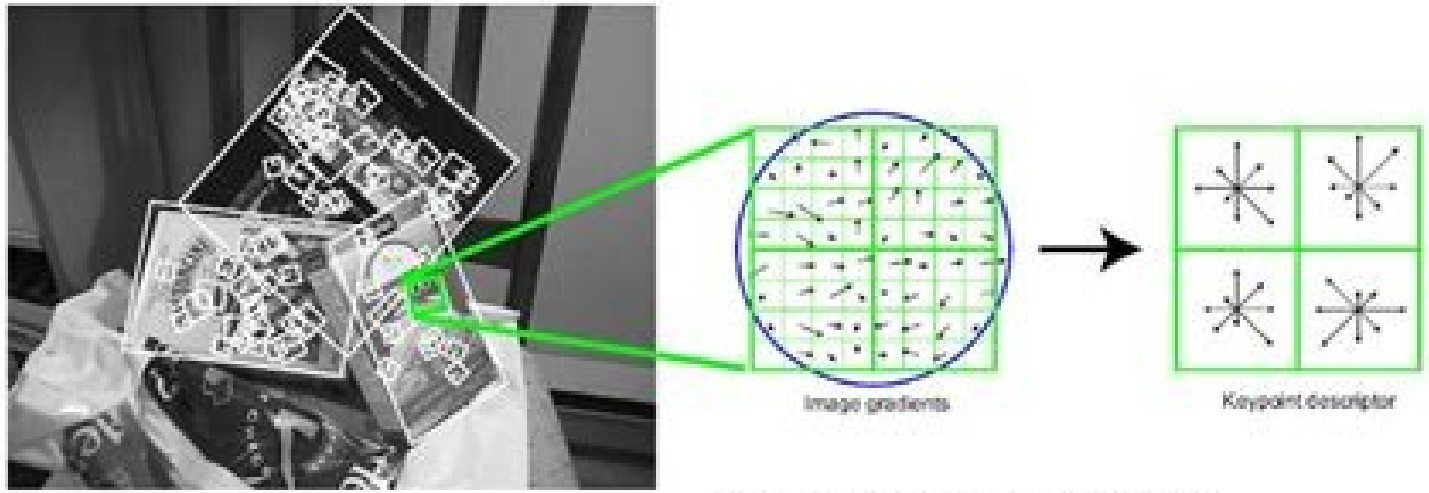
Limitations

Feature-based methods required domain expertise and struggled with variations in lighting, viewpoint, and occlusion.



Scale-Invariant Feature Transform (SIFT) descriptor

Journal + conference versions: 70,015 citations – still the most influential vision work



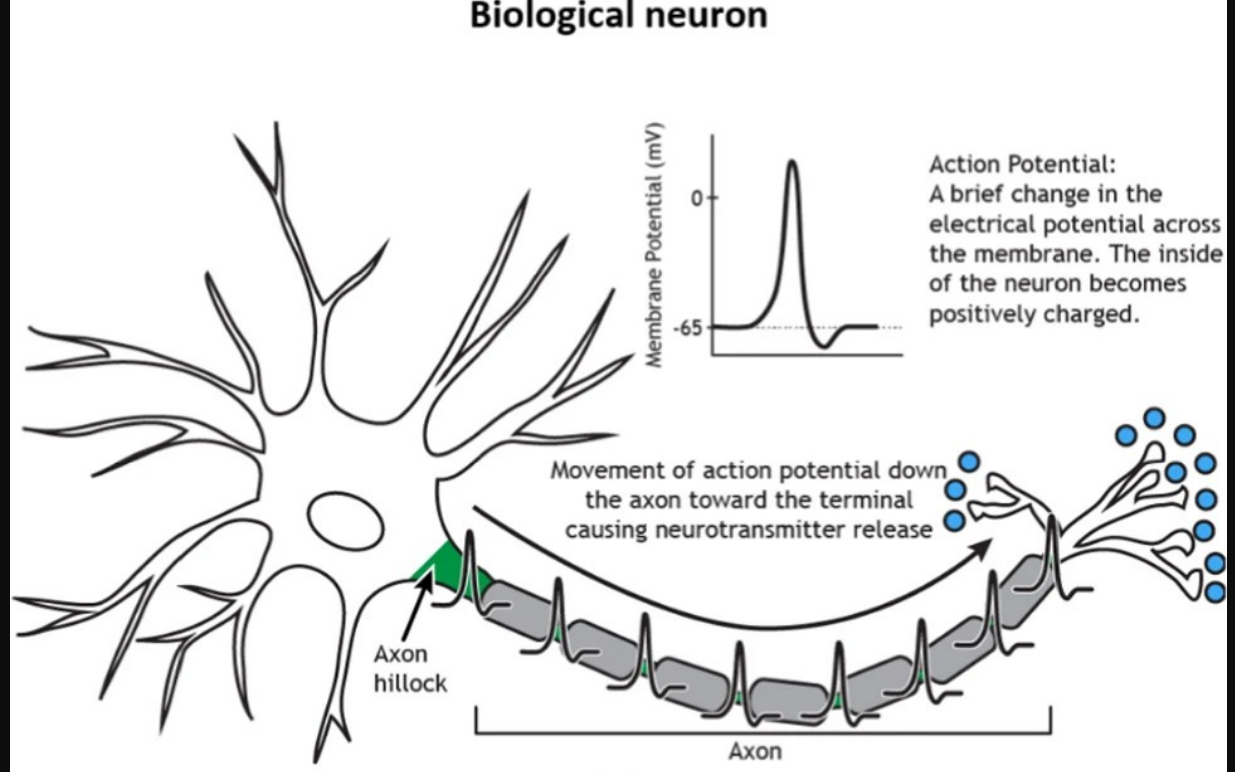
Histogram of oriented gradients

- **Captures important texture information**
- **Robust to small translations / affine deformations**

[Lowe, ICCV 1999]

K. Grauman, B. Leibe

Machine Learning Bio-inspired perceptron



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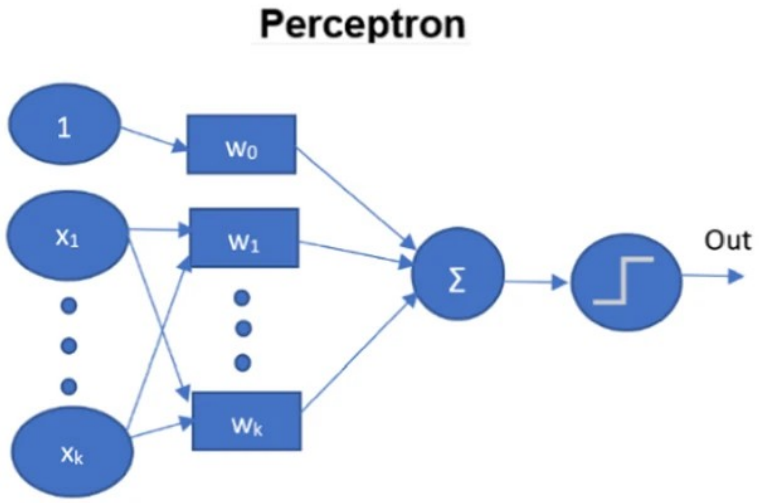
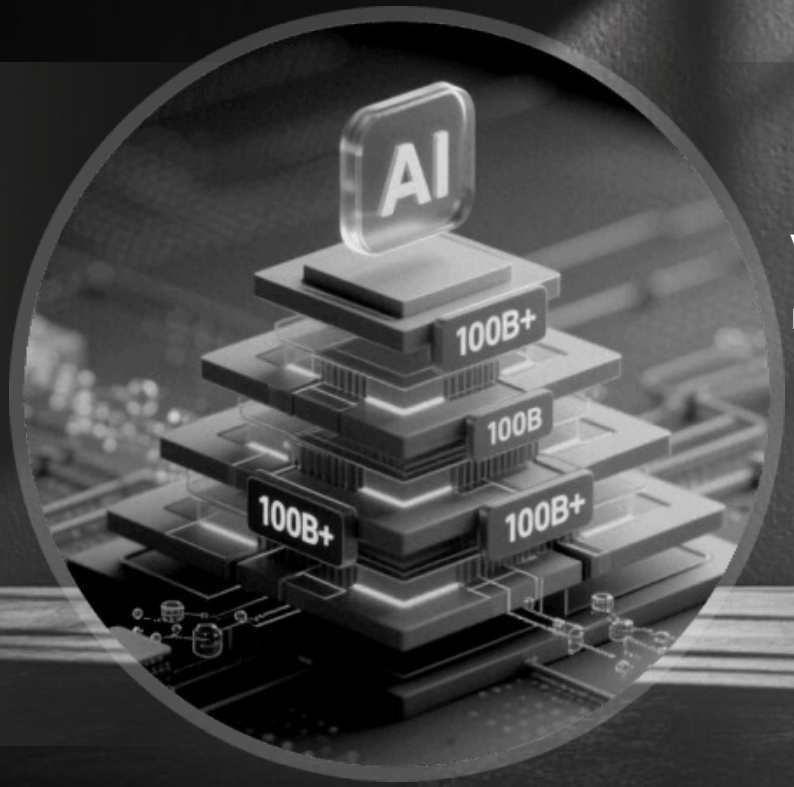


(b)

Machine Learning Revolution 2000s

Statistical Learning

Machine learning techniques like boosting, SVMs, and random forests replace fixed rules.



Impact

Viola & Jones cascaded detectors enabled real-time face detection on CPUs.

Deep Learning Breakthrough 2012

01

AlexNet's Victory

AlexNet's GPU-trained CNN won the 2012 ImageNet competition, proving that hierarchical learned features outperform engineered ones.

02

Architectural Innovations

Deeper networks like VGG, Inception, and ResNet, which achieved state-of-the-art performance on various vision tasks.

03

Future Directions

The deep learning revolution continues to drive advancements in computer vision, with ongoing research exploring new architectures, training techniques, and applications.

Eye to Cortex Pathway

01

Light enters the eye and is converted into electrical signals by photoreceptors. These signals finally reach the visual cortex, where the are processed hierarchically.

Retinotopic Mapping

02

The visual pathway preserves spatial layout through retinotopic maps, enabling parallel processing of features across scales.

Hierarchical Processing

03

The hierarchical structure of the visual cortex, with different regions specializing in various aspects of visual processing, has inspired the design of deep learning models like CNNs.

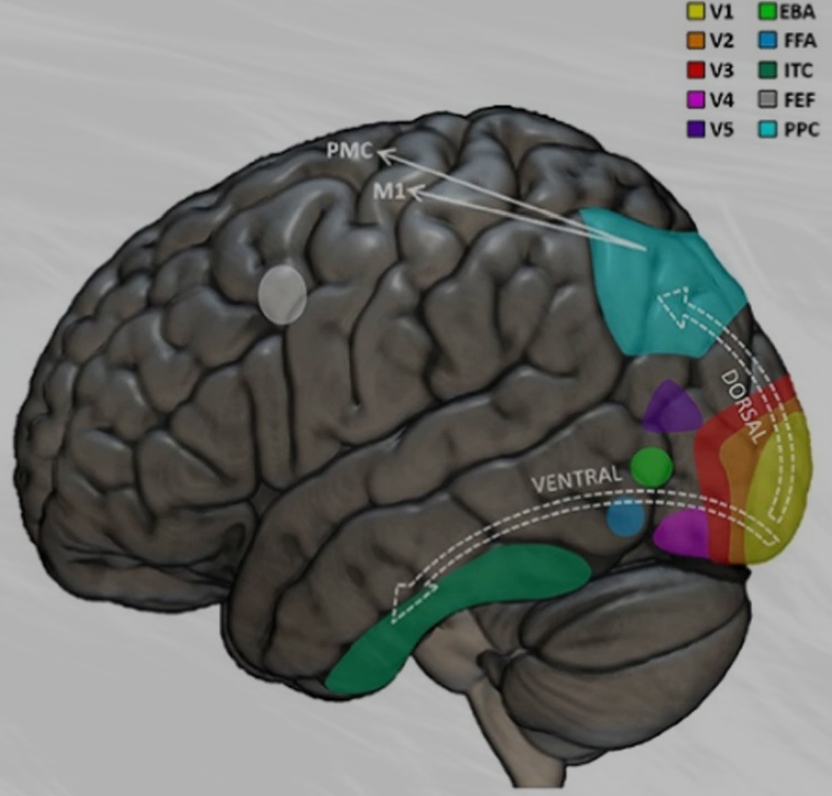
Ventral and Dorsal Streams

Two-stream hypothesis proposes that visual information is processed along two pathways: the ventral stream for object recognition and the dorsal stream for spatial awareness and motion perception.

Functional Specialization

The ventral stream is responsible for identifying objects and scenes. The dorsal stream handles spatial information and guides actions.

Eye to Cortex Pathway



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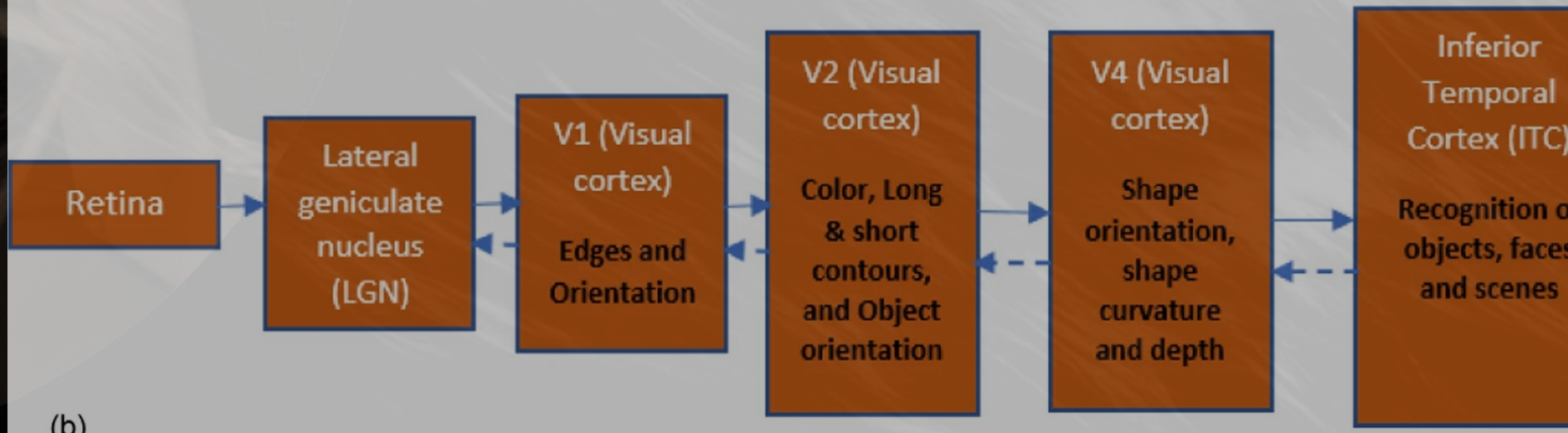

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Two-Stream Hypothesis

Ventral and Dorsal Streams

The two-stream hypothesis proposes that visual information is processed along two distinct pathways: the ventral stream for object recognition and the dorsal stream for spatial awareness and motion perception.

Functional Specialization

The ventral stream, known as the 'what' pathway, is responsible for identifying objects and scenes. The dorsal stream, or 'where' pathway, handles spatial information and guides actions.

Parallelism & Hierarchy

01

Parallel Processing

The human visual system processes information in parallel, with millions of neurons working simultaneously to analyze different aspects of the visual scene. This allows for rapid and efficient interpretation of complex visual data.

Hierarchical Organization

02

The hierarchical structure of the visual system, with each stage building on the output of the previous one, enables the extraction of increasingly complex and abstract features from the visual input.

03

Inspiration for CNNs

The principles of parallelism and hierarchy in the human visual system have inspired the design of Convolutional Neural Networks, which mimic this structure to achieve state-of-the-art performance in computer vision tasks.

Recurrence is Essential

Cortical Feedback

Modern neuroscience reveals that cortical feedback fibers carry top-down predictions, enabling noise suppression and occlusion completion. Recurrent CNNs now outperform feed-forward models on robustness benchmarks.

Advantages

Recurrent connections allow for dynamic, context-aware processing, improving robustness to noise and occlusion. This aligns with how the human brain processes visual information.

Applications

Recurrent models are particularly useful in tasks requiring temporal consistency and context, such as video analysis and autonomous navigation.

Architectures

Convolutional Neural Networks

CNNs are designed for processing grid-like data, such as images, using convolutional and pooling layers to extract hierarchical features. They form the foundation of modern computer vision.

Recurrent Models for Video

RNNs, including LSTM and GRU, process sequential data by maintaining a memory of previous inputs. This makes them suitable for tasks involving temporal dependencies, such as video analysis.

Vision Transformers

Vision Transformers treat images as sequences of patches, using self-attention mechanisms to capture global context. This approach has shown promise in various vision tasks.

Bio-Inspired CNN Variants

Biological Inspiration	Bio-inspired CNN variants incorporate principles from the human visual system, such as Gabor filters and recurrent connections, to improve robustness and efficiency.
Advantages	These models often achieve better performance with fewer parameters, making them more efficient and robust to variations in lighting, noise, and other conditions.
Applications	Bio-inspired models are particularly useful in applications where interpretability and robustness are critical, such as medical imaging and autonomous systems.

Explainable AI Techniques

Techniques

Saliency maps, Grad-CAM (Gradient-weighted Class Activation Mapping), and LRP (Layer-wise Relevance Propagation) visualize input regions influencing model predictions, enhancing interpretability and trust in AI systems.

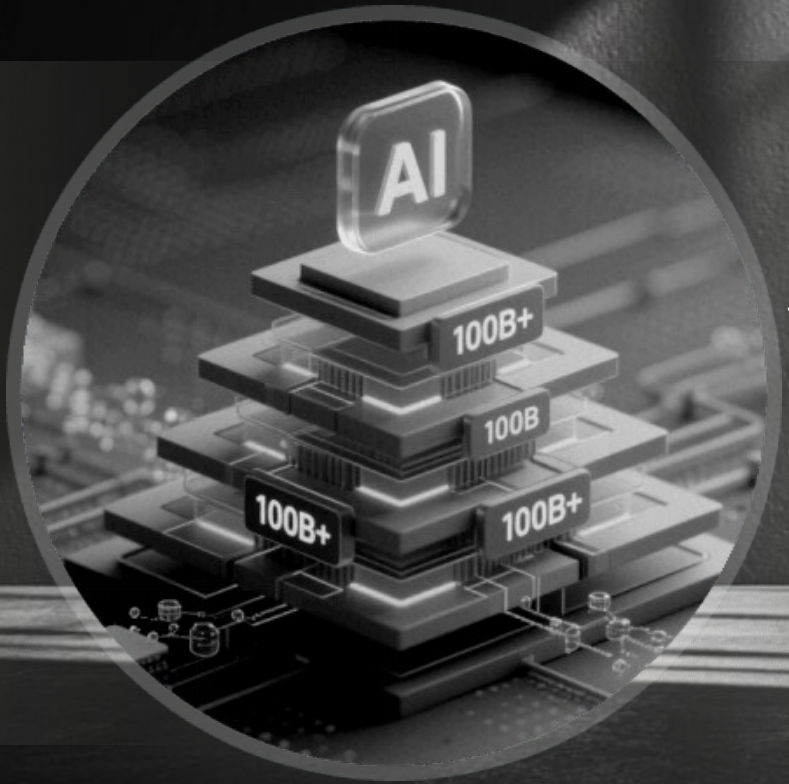
Applications

Explainable AI techniques are crucial in high-stakes applications like medical diagnosis and autonomous driving, where understanding model decisions is essential.

Algorithmic Bias Consequences

Bias Issues

Skewed training data can propagate racial, gender, and socioeconomic disparities in applications like hiring, policing, and lending.



Mitigation

Continuous auditing, diverse teams, and transparent reporting are necessary to detect and mitigate bias, ensuring equitable outcomes.

Hybrid Bio-Inspired Path

Hybrid Models

Next-generation vision systems will blend CNN representation power with recurrent feedback, spike-based efficiency, and self-supervised learning, approaching biological flexibility and energy efficiency.

Future Work

Research will focus on integrating principles like saccadic mechanisms, foveated attention, and lifelong plasticity to create more advanced vision systems.

Advantages

These hybrid models will offer improved robustness, adaptability, and efficiency, making them suitable for a wide range of applications from healthcare to robotics.

Thank you.